Lecture 16

The Weyl character formula

Objectives. *A Verma character is an infinite series. Let $\lambda \in P_+$. We prove that the product of the Weyl denominator and the character of $L(\lambda)$ is a single finite alternant. The Casimir element puts the exponents of this product on a sphere. Dominance then leaves one Weyl orbit.*

By the end of this lecture you will be able to:

- Derive the Casimir scalar $(\mu, \mu + 2\rho)$ on a highest-weight module of highest weight μ.
- Write $\operatorname{ch} L(\lambda)$ as a finite integer combination of Verma characters.
- Prove the Weyl character formula for $\lambda \in P_+$ from the norm and dominance conditions.
- Display the cancellations for $\mathfrak{sl}_2$ and for the adjoint module of $\mathfrak{sl}_3$.

Throughout, $\mathfrak{g}$ is a nonzero complex semisimple Lie algebra with the data $\mathfrak{h}$, R_+, $\alpha_1, \dots, \alpha_r$, W, Q, P, P_+ and ρ of Lecture 9. Highest-weight modules are infinite-dimensional in general. This is a departure from the standing convention that modules are finite-dimensional. The final theorem concerns only the finite-dimensional modules $L(\lambda)$, $\lambda \in P_+$. We write

$$Q_+ = \sum_{i=1}^{r} \mathbb{Z}_{\geq 0}\alpha_i, \qquad \mu \leq \nu \iff \nu - \mu \in Q_+.$$

Characters lie in the completion of Lecture 15. Its elements are formal sums $\sum c_\nu e^\nu$ supported in a finite union of sets $\xi - Q_+$ with $\xi \in P$. For $\nu \in P$ define the alternant A_ν and the Weyl denominator D:

$$A_\nu = \sum_{w \in W} \det(w) e^{w\nu}, \qquad D = e^\rho \prod_{\alpha \in R_+} (1 - e^{-\alpha}).$$

From Lecture 15 we use the Weyl invariance of finite-dimensional characters and the Verma character: for $\mu \in P$,

$$\operatorname{ch} M(\mu) \prod_{\alpha \in R_+} (1 - e^{-\alpha}) = e^\mu.$$

Multiplying this identity by e^ρ gives

$$D \operatorname{ch} M(\mu) = e^{\mu + \rho}. \tag{16.1}$$

From Lecture 15 we also use the denominator identity $D = A_\rho$.

For $w \in W$, reindex by $u' = wu$. Then $\det(u) = \det(w)\det(u')$, so

$$\begin{aligned} wA_\nu &= \sum_{u \in W} \det(u) e^{wu\nu} = \det(w) \sum_{u' \in W} \det(u') e^{u'\nu} = \det(w) A_\nu, \\ wD &= \det(w) D. \end{aligned} \tag{16.2}$$

The last identity uses $D = A_\rho$.

16.1 The Casimir element of $U(\mathfrak{g})$

Fix the invariant symmetric form $(\cdot,\cdot)$ on $\mathfrak{g}$ of the course. This form is nondegenerate on $\mathfrak{g}$. On each simple ideal it is normalised so that long roots have squared length 2. Distinct simple ideals are orthogonal. On $\mathfrak{sl}_n$ it is $\operatorname{tr}(XY)$, while the Killing form is $2n \operatorname{tr}(XY)$.

The restriction of the form to $\mathfrak{h}$ is nondegenerate (Lecture 7). Hence for $\mu \in \mathfrak{h}^*$ there is a unique $t_\mu \in \mathfrak{h}$ with

$$(t_\mu, H) = \mu(H) \quad \text{for all } H \in \mathfrak{h}.$$

On $\mathfrak{h}^*$ put

$$(\mu, \nu) = (t_\mu, t_\nu) = \mu(t_\nu), \qquad \|\mu\|^2 = (\mu, \mu).$$

On the real span of the roots this form is W-invariant and positive definite (Lecture 8, Lecture 9).

Lemma 16.1 (Casimir element)**.** *Let (x_i) be a basis of $\mathfrak{g}$ and (x^i) the dual basis, so that $(x_i, x^j) = \delta_{ij}$. Then the element*

$$\Omega = \sum_i x_i x^i \in U(\mathfrak{g})$$

is central in $U(\mathfrak{g})$.

Proof. Fix $y \in \mathfrak{g}$. Expanding in the two bases gives

$$[y, x_i] = \sum_j c_{ji} x_j, \qquad c_{ji} = ([y, x_i], x^j).$$

The coefficient of x^j in $[y, x^i]$ is $([y, x^i], x_j)$. Invariance gives

$$([y, x^i], x_j) = -(x^i, [y, x_j]) = -\Big(x^i, \sum_k c_{kj} x_k\Big) = -c_{ij},$$

so $[y, x^i] = -\sum_j c_{ij} x^j$. In $U(\mathfrak{g})$, the map $u \mapsto yu - uy$ obeys the Leibniz rule. Hence

$$\begin{aligned} y\Omega - \Omega y &= \sum_i \big([y, x_i] x^i + x_i [y, x^i] \big) \\ &= \sum_{i,j} c_{ji} x_j x^i - \sum_{i,j} c_{ij} x_i x^j = 0. \end{aligned}$$

The last step exchanges i and j in the second sum. Since $\mathfrak{g}$ generates $U(\mathfrak{g})$, the element Ω is central. ∎

The change-of-basis calculation of Lecture 5, followed by multiplication into $U(\mathfrak{g})$, shows that Ω is basis-independent. Now choose a basis adapted to the root space decomposition. Root spaces are one-dimensional (Lecture 8). The form pairs $\mathfrak{g}_\alpha$ with $\mathfrak{g}_{-\alpha}$ nondegenerately.

It vanishes on all other pairs of root spaces and between $\mathfrak{h}$ and root spaces (Lecture 7). For $\alpha \in R_+$ choose $e_\alpha \in \mathfrak{g}_\alpha$ and $f_\alpha \in \mathfrak{g}_{-\alpha}$ with $(e_\alpha, f_\alpha) = 1$. Choose an orthonormal basis $H_1, \dots, H_r$ of $\mathfrak{h}$. It exists because the form on $\mathfrak{h}$ is symmetric and nondegenerate and the field is $\mathbb{C}$. The dual basis exchanges e_α and f_α and fixes each H_i. Hence

$$\Omega = \sum_{i=1}^{r} H_i^2 + \sum_{\alpha \in R_+} (e_\alpha f_\alpha + f_\alpha e_\alpha).$$

The bracket $[e_\alpha, f_\alpha]$ lies in $\mathfrak{g}_0 = \mathfrak{h}$. For $H \in \mathfrak{h}$,

$$(H, [e_\alpha, f_\alpha]) = ([H, e_\alpha], f_\alpha) = \alpha(H)(e_\alpha, f_\alpha) = \alpha(H).$$

Thus $[e_\alpha, f_\alpha] = t_\alpha$, which gives the first identity below. The second uses $\sum_{\alpha \in R_+} \alpha = 2\rho$ and the linearity of $\mu \mapsto t_\mu$:

$$e_\alpha f_\alpha = f_\alpha e_\alpha + t_\alpha, \qquad \sum_{\alpha \in R_+} t_\alpha = t_{2\rho}.$$

Substituting both identities into Ω gives

$$\Omega = \sum_{i=1}^{r} H_i^2 + t_{2\rho} + 2 \sum_{\alpha \in R_+} f_\alpha e_\alpha. \tag{16.3}$$

Definition 16.2 (Highest-weight module)**.** Let $\mu \in \mathfrak{h}^*$. A $\mathfrak{g}$-module U is a *highest-weight module of highest weight* μ if it is generated by a vector $v \neq 0$ with

$$e_\alpha v = 0 \ (\alpha \in R_+), \qquad Hv = \mu(H)v \ (H \in \mathfrak{h}).$$

Use $\mathfrak{n}_-$ from Lecture 11. By Lecture 12, such a module satisfies $U = U(\mathfrak{n}_-)v$. The module U is the direct sum of finite-dimensional weight spaces U_ν with $\nu \leq \mu$, and $U_\mu = \mathbb{C}v$. The Verma module $M(\mu)$ is one example. An irreducible highest-weight module of highest weight μ is isomorphic to $L(\mu)$.

Proposition 16.3 (Casimir scalar)**.** *On every highest-weight module U of highest weight μ,*

$$\Omega = (\mu, \mu + 2\rho)\ \mathrm{id}_U = \big(\|\mu + \rho\|^2 - \|\rho\|^2\big)\ \mathrm{id}_U\,.$$

Proof. Expand t_μ in the orthonormal basis:

$$t_\mu = \sum_i (t_\mu, H_i) H_i = \sum_i \mu(H_i) H_i, \qquad (\mu, \mu) = \mu(t_\mu) = \sum_i \mu(H_i)^2.$$

Apply (16.3) to v. Each e_α kills v, so the last sum vanishes:

$$\Omega v = \Big(\sum_i \mu(H_i)^2 + \mu(t_{2\rho})\Big) v = \big((\mu, \mu) + (\mu, 2\rho)\big) v = (\mu, \mu + 2\rho) v.$$

Every vector of U has the form uv with $u \in U(\mathfrak{g})$. Centrality gives

$$\Omega(uv) = u\,\Omega v = (\mu, \mu + 2\rho)\, uv.$$

∎

Replacing the form by $c \neq 0$ times itself divides Ω and its scalars by c. Exercise 16.1 compares the trace and Killing forms.

16.2 Highest-weight modules on one Casimir sphere

Lemma 16.4 (Weight submodules). *Let $V = \bigoplus_{\nu \in P} V_\nu$ be a $\mathfrak{g}$-module with finite-dimensional weight spaces. Suppose all weights lie in a finite union of sets $\xi - Q_+$ with $\xi \in P$. Highest-weight modules with highest weight in P are examples. Let $X \subseteq V$ be a submodule. Then $X = \bigoplus_\nu (X \cap V_\nu)$. Moreover, X and V/X satisfy the same hypotheses, so all three characters are defined, and*

$$\operatorname{ch} V = \operatorname{ch} X + \operatorname{ch}(V/X).$$

Proof. Let $u = v_1 + \cdots + v_k \in X$ with nonzero $v_j \in V_{\nu_j}$ and distinct ν_j. For $i \neq j$, the kernel of $\nu_i - \nu_j$ is a proper subspace of $\mathfrak{h}$. A complex vector space is not a finite union of proper subspaces. So some $H \in \mathfrak{h}$ has pairwise distinct values $\nu_j(H)$. Then

$$\prod_{j=2}^{k} \big(H - \nu_j(H)\big) u = \prod_{j=2}^{k} \big(\nu_1(H) - \nu_j(H)\big) v_1 \in X.$$

The scalar on the right is nonzero, so $v_1 \in X$. Likewise every v_j lies in X.

Hence V/X is the direct sum of the images of the V_ν. With $X_\nu = X \cap V_\nu$, the image of V_ν is V_ν/X_ν. It has weight ν, and

$$\dim (V/X)_\nu = \dim V_\nu - \dim X_\nu.$$

This is the character identity, coefficient by coefficient. ∎

Fix $\lambda \in P$ and define the *Casimir sphere*

$$S_\lambda = \{\mu \in P : \|\mu + \rho\|^2 = \|\lambda + \rho\|^2\}.$$

The basis $\omega_1, \dots, \omega_r$ makes P a lattice in a Euclidean space. A sphere contains only finitely many lattice points, so S_λ is finite. Subtracting $\|\rho\|^2$ from both sides of the defining condition gives

$$\mu \in S_\lambda \iff \mu \in P \text{ and } (\mu, \mu + 2\rho) = (\lambda, \lambda + 2\rho).$$

For $\mu \in P$, Proposition 16.3 gives $\mu \in S_\lambda$ exactly when Ω has the same scalar on $M(\mu)$ as on $M(\lambda)$.

For a highest-weight module U with highest weight in S_λ, set

$$N(U) = \sum_{\nu \in S_\lambda} \dim U_\nu.$$

It is finite, and $N(U) \geq 1$ because the highest weight contributes.

Proposition 16.5 (Decomposition on the sphere). *Let U be a highest-weight module of highest weight $\mu \in S_\lambda$. Then*

$$\operatorname{ch} U = \sum_{\nu \in S_\lambda,\ \nu \leq \mu} a_\nu \operatorname{ch} L(\nu), \qquad a_\nu \in \mathbb{Z}_{\geq 0}, \quad a_\mu = 1.$$

Proof. We use strong induction on $N(U)$. If U is irreducible, then $U \cong L(\mu)$ and there is nothing to prove.

Step 1: a new highest-weight vector. Let X be a proper nonzero submodule. By Lemma 16.4, X is a sum of weight spaces. Also $X_\mu = 0$. Otherwise X would contain $U_\mu = \mathbb{C}v$, and then $X = U$. Pick a weight ν_0 of X. Consider the set

$$E = \{\xi : X_\xi \neq 0,\ \nu_0 \leq \xi \leq \mu\}.$$

It contains ν_0. It is finite, since $\xi - \nu_0 \in Q_+$ and only finitely many elements of Q_+ lie below $\mu - \nu_0$. Let η be maximal in E. All weights of U are $\le \mu$, so every weight $\xi > \eta$ of X would lie in E. Hence η is maximal among all weights of X. Also $\eta < \mu$, because $X_\mu = 0$. Take a nonzero vector $x \in X_\eta$. By the maximality of η, for every $\alpha \in R_+$,

$$e_\alpha x \in X_{\eta+\alpha} = 0.$$

Step 2: the Casimir. Since x is a weight vector annihilated by all positive root vectors, the calculation of Proposition 16.3 applies to x. Also $x \in U$, so

$$(\eta, \eta + 2\rho)x = \Omega x = (\mu, \mu + 2\rho)x.$$

As $\eta \in \mu - Q \subseteq P$, this gives $\eta \in S_\lambda$.

Step 3: induction. The submodule $X' = U(\mathfrak{g})x$ is a highest-weight module of highest weight η, and $X'_\mu = 0$ since $X' \subseteq X$. The quotient U/X' is a highest-weight module of highest weight μ, generated by the image of v. Lemma 16.4 gives

$$\operatorname{ch} U = \operatorname{ch} X' + \operatorname{ch}(U/X'), \qquad N(U) = N(X') + N(U/X').$$

Both summands of $N(U)$ are at least 1, so both are smaller than $N(U)$. By induction, $\operatorname{ch} X'$ involves only $L(\nu)$ with $\nu \le \eta < \mu$, and $\operatorname{ch}(U/X')$ contains $\operatorname{ch} L(\mu)$ exactly once. Adding the two expansions proves the claim. ∎

16.3 Triangular inversion and candidate weights

Enumerate $S_\lambda = \{\mu_1, \dots, \mu_l\}$ so that $\mu_i < \mu_j$ implies $i < j$. To do this, repeatedly remove a minimal element. Apply Proposition 16.5 to $U = M(\mu_k)$:

$$\operatorname{ch} M(\mu_k) = \operatorname{ch} L(\mu_k) + \sum_{j<k} a_{kj} \operatorname{ch} L(\mu_j). \tag{16.4}$$

Set $a_{kk} = 1$ and $a_{kj} = 0$ for $j > k$. Then $A = (a_{kj})$ is lower unitriangular with integer entries.

Proposition 16.6 (Inversion). *For each k there are integers b_{kj} with $b_{kk} = 1$ and*

$$\operatorname{ch} L(\mu_k) = \sum_{j \le k} b_{kj} \operatorname{ch} M(\mu_j).$$

Proof. Use induction on k. Rearranging (16.4) gives

$$\operatorname{ch} L(\mu_k) = \operatorname{ch} M(\mu_k) - \sum_{j<k} a_{kj} \operatorname{ch} L(\mu_j).$$

By induction, each $\operatorname{ch} L(\mu_j)$ with $j < k$ is an integer combination of $\operatorname{ch} M(\mu_i)$ with $i \le j$. Substituting proves the claim. ∎

Theorem 16.7 (Finite candidate weights). *Let $\lambda \in P_+$. There are integers b_μ such that*

$$D \operatorname{ch} L(\lambda) = \sum_{\mu \in S_\lambda} b_\mu e^{\mu+\rho}.$$

Every exponent ν with nonzero coefficient satisfies $\|\nu\|^2 = \|\lambda + \rho\|^2$.

Proof. Since $\lambda \in S_\lambda$, Proposition 16.6 writes $\operatorname{ch} L(\lambda)$ as an integer combination of $\operatorname{ch} M(\mu)$, $\mu \in S_\lambda$. Multiply by D and apply (16.1). The exponents $\mu + \rho$ are distinct and lie on the stated sphere. ∎

Both sides lie in $\mathbb{Z}[P]$, since $L(\lambda)$ is finite-dimensional (Lecture 13).

16.4 Eliminating lower alternants

Theorem 16.8 (Weyl character formula)**.** *Let $\lambda \in P_+$. Then $D \operatorname{ch} L(\lambda) = A_{\lambda+\rho}$. Since $D = A_\rho$,*

$$\operatorname{ch} L(\lambda) = \frac{A_{\lambda+\rho}}{A_\rho}, \qquad A_\nu = \sum_{w \in W} \det(w) e^{w\nu}.$$

Proof. Write $F = D \operatorname{ch} L(\lambda) = \sum_\nu c_\nu e^\nu$.

(a) Norm. If $c_\nu \neq 0$, then $\|\nu\|^2 = \|\lambda + \rho\|^2$, by Theorem 16.7.

(b) Dominance bound. Expand the product D:

$$D = e^\rho \prod_{\alpha \in R_+} (1 - e^{-\alpha}) = \sum_{T \subseteq R_+} (-1)^{|T|} e^{\rho - \sum_{\alpha \in T} \alpha}.$$

Each exponent of F therefore has the form

$$\nu = \mu + \rho - \sum_{\alpha \in T} \alpha, \qquad \mu \text{ a weight of } L(\lambda),\ T \subseteq R_+.$$

Since $\lambda - \mu$ and $\sum_{\alpha \in T} \alpha$ both lie in Q_+, every such ν satisfies $\nu \leq \lambda + \rho$. Equality forces $\mu = \lambda$ and $T = \emptyset$, because a nonempty sum of positive roots is a nonzero element of Q_+. As $\dim L(\lambda)_\lambda = 1$, we get $c_{\lambda+\rho} = 1$.

(c) Alternation. The character is W-invariant (Lecture 15), and $wD = \det(w) D$ by (16.2). Since w acts on $\mathbb{Z}[P]$ by ring automorphisms,

$$wF = (wD)\,(w \operatorname{ch} L(\lambda)) = \det(w) D \operatorname{ch} L(\lambda) = \det(w) F.$$

Comparing coefficients of $e^{w\nu}$ gives $c_{w\nu} = \det(w) c_\nu$.

(d) Norm and dominance together. Let $c_\nu \neq 0$. By Lecture 9, $\nu' = w^{-1}\nu$ is dominant for some $w \in W$. By (c), $c_{\nu'} = \pm c_\nu \neq 0$, so (a) and (b) apply to ν'. By (b), the element $\gamma = \lambda + \rho - \nu'$ lies in Q_+. Then

$$\begin{aligned} \|\lambda + \rho\|^2 = \|\nu' + \gamma\|^2 &= \|\nu'\|^2 + 2(\nu', \gamma) + \|\gamma\|^2, \\ 0 &= 2(\nu', \gamma) + \|\gamma\|^2. \end{aligned}$$

The last line uses (a). Write $\gamma = \sum_i k_i \alpha_i$ with $k_i \geq 0$. Dominance gives $(\nu', \alpha_i) \geq 0$, hence

$$(\nu', \gamma) = \sum_i k_i (\nu', \alpha_i) \geq 0, \qquad \|\gamma\|^2 = -2(\nu', \gamma) \leq 0.$$

Positive definiteness forces $\gamma = 0$. Thus $\nu' = \lambda + \rho$ and $\nu = w(\lambda + \rho)$.

(e) Conclusion. By (b) and (c), $c_{w(\lambda+\rho)} = \det(w)$. If $w \neq 1$, some $\alpha \in R_+$ has $w\alpha \in -R_+$ (Lecture 9). For each simple root, $(\lambda, \alpha_i) \geq 0$ and $(\rho, \alpha_i) = \frac{1}{2}(\alpha_i, \alpha_i) > 0$. Every $\beta \in R_+$ is a nonzero combination of simple roots with nonnegative coefficients. Hence $(\lambda + \rho, \beta) > 0$ for all $\beta \in R_+$, and

$$(w(\lambda + \rho), w\alpha) = (\lambda + \rho, \alpha) > 0 > (\lambda + \rho, w\alpha).$$

So $w(\lambda + \rho) \neq \lambda + \rho$. Applying this to $u^{-1}w$ for $u \neq w$ shows that the points $w(\lambda + \rho)$ are distinct. By (d),

$$F = \sum_{w \in W} c_{w(\lambda+\rho)} e^{w(\lambda+\rho)} = \sum_{w \in W} \det(w) e^{w(\lambda+\rho)} = A_{\lambda+\rho}.$$

Finally, $\mathbb{Z}[P]$ is a Laurent polynomial ring in $e^{\omega_1}, \ldots, e^{\omega_r}$, hence an integral domain. By the expansion of D in (b), $A_\rho = D$ has top coefficient 1, so $A_\rho \neq 0$. The quotient is the unique element whose product with A_ρ is $A_{\lambda+\rho}$. ∎

16.5 Rank one and the adjoint module of $\mathfrak{sl}_3$

Example 16.9 ($\mathfrak{sl}_2$)**.** Here $\rho = \omega = \alpha/2$, $(\omega,\omega) = \frac{1}{2}$ and $W = \{1, s\}$. Write $x = e^{\omega}$ and $\lambda = m\omega$, with $m \in \mathbb{Z}_{\geq 0}$. For $\mu = k\omega$,

$$\|\mu+\rho\|^2 = \frac{1}{2}(k+1)^2, \qquad \frac{1}{2}(k+1)^2 = \frac{1}{2}(m+1)^2 \iff k \in \{m, -m-2\}.$$

Hence

$$S_\lambda = \{m\omega,\ -(m+2)\omega\}.$$

Exercise 16.2 proves the Verma expansion on this sphere:

$$\operatorname{ch} L(m\omega) = \operatorname{ch} M(m\omega) - \operatorname{ch} M(-(m+2)\omega).$$

Multiply by $D = A_\rho = x - x^{-1}$ and use (16.1):

$$A_\rho \operatorname{ch} L(m\omega) = e^{(m+1)\omega} - e^{-(m+1)\omega} = x^{m+1} - x^{-m-1}.$$

The cancellations are also visible directly. For $m = 0$ both sides equal $x - x^{-1}$. For $m \geq 1$,

$$\begin{aligned}(x - x^{-1}) \sum_{k=0}^{m} x^{m-2k} &= (x^{m+1} + x^{m-1} + \cdots + x^{-m+1}) \\ &\quad - (x^{m-1} + \cdots + x^{-m+1} + x^{-m-1}) \\ &= x^{m+1} - x^{-m-1}.\end{aligned}$$

Example 16.10 (The adjoint module of $\mathfrak{sl}_3$)**.** Here $\theta = \alpha_1 + \alpha_2 = \rho$ and $\|\rho\|^2 = 2$. For $w = 1, s_1, s_2, s_1s_2, s_2s_1, w_0$, with $w_0 = s_1s_2s_1$, the points $w\rho$ are, in this order,

$$\alpha_1+\alpha_2, \quad \alpha_2, \quad \alpha_1, \quad -\alpha_1, \quad -\alpha_2, \quad -\alpha_1-\alpha_2.$$

Hence

$$\begin{aligned}A_\rho &= e^{\alpha_1+\alpha_2} - e^{\alpha_2} - e^{\alpha_1} + e^{-\alpha_1} + e^{-\alpha_2} - e^{-\alpha_1-\alpha_2}, \\ A_{2\rho} &= e^{2\alpha_1+2\alpha_2} - e^{2\alpha_2} - e^{2\alpha_1} + e^{-2\alpha_1} + e^{-2\alpha_2} - e^{-2\alpha_1-2\alpha_2}.\end{aligned}$$

The adjoint module is $L(\theta)$. Its nonzero weights are the six roots, each with multiplicity one. We assume only this and let the formula determine the multiplicity m of the weight 0. Set

$$\chi_m = \sum_{\alpha\in R} e^{\alpha} + m\, e^{0}.$$

Every χ_m is W-invariant, so $A_\rho\chi_m$ is alternating, as in step (c). Its coefficients on a W-orbit are fixed by the dominant representative. Every weight of χ_m is $\leq \theta$. So the argument of (b), applied to $A_\rho\chi_m$, bounds all exponents by 2ρ. Since $\omega_1 = \frac{1}{3}(2\alpha_1+\alpha_2)$ and $\omega_2 = \frac{1}{3}(\alpha_1+2\alpha_2)$, the dominant elements of Q below 2ρ are those in the first column:

Exponent	Products (term of A_ρ)·(term of χ_m)	Coefficient
$2\alpha_1+2\alpha_2$	$e^{\rho}\cdot e^{\theta}$	1
$2\alpha_1+\alpha_2$	$e^{\rho}\cdot e^{\alpha_1}$, $-e^{\alpha_1}\cdot e^{\theta}$	0
$\alpha_1+2\alpha_2$	$e^{\rho}\cdot e^{\alpha_2}$, $-e^{\alpha_2}\cdot e^{\theta}$	0
$\alpha_1+\alpha_2$	$e^{\rho}\cdot me^{0}$, $-e^{\alpha_1}\cdot e^{\alpha_2}$, $-e^{\alpha_2}\cdot e^{\alpha_1}$	$m-2$
0	$\det(w)e^{w\rho}\cdot e^{-w\rho}$, $w\in W$	$3-3=0$

Therefore

$$A_\rho \chi_m = A_{2\rho} + (m-2)A_\rho.$$

The surplus $(m-2)A_\rho$ is a lower alternant. Alternation (c) and the top coefficient (b) cannot remove it. The norm condition (a) does. Using the Gram matrix in Lecture 9, for $\nu = a\omega_1 + b\omega_2$,

$$\|\nu\|^2 = \frac{2}{3}(a^2+ab+b^2), \qquad \|2\rho\|^2 = \frac{2}{3}(4+4+4) = 8.$$

The condition $\|\nu\|^2 = 8$ reads $a^2+ab+b^2 = 12$, while $\|\rho\|^2 = 2 \neq 8$. Theorem 16.8 therefore forces $m = 2$:

$$\dim L(\theta)_0 = 2 = \dim \mathfrak{h}, \qquad \dim L(\theta) = 6+2 = 8.$$

Self-checks

(1) What is the Casimir scalar on the adjoint module of $\mathfrak{sl}_3$, for $\operatorname{tr}(XY)$ and for the Killing form?

(2) For $\mathfrak{sl}_2$ and $\lambda = 3\omega$, what are S_λ and the Verma expansion of $\operatorname{ch} L(\lambda)$?

(3) Why does e^0 never occur in $A_\rho \operatorname{ch} L(\lambda)$?

Answers. (1) $(\theta, \theta+2\rho) = (\rho, 3\rho) = 6$. The Killing form is $6\operatorname{tr}(XY)$, so its scalar is 1. (2) $S_\lambda = \{3\omega, -5\omega\}$ and $\operatorname{ch} L(3\omega) = \operatorname{ch} M(3\omega) - \operatorname{ch} M(-5\omega)$. (3) Since $\mathfrak{g} \neq 0$, there is a simple reflection s_1. It fixes 0, so (c) gives $c_0 = -c_0$.

Summary

The Casimir element acts on a highest-weight module of highest weight μ by $\|\mu+\rho\|^2 - \|\rho\|^2$. The Verma characters with highest weights on a finite Casimir sphere are unitriangular combinations of irreducible characters. Let $\lambda \in P_+$. Inverting the triangular system puts the exponents of $A_\rho \operatorname{ch} L(\lambda)$ on the sphere. Dominance and the norm leave only $W(\lambda+\rho)$, and this gives $A_\rho \operatorname{ch} L(\lambda) = A_{\lambda+\rho}$. Lecture 17 turns the formula into dimensions and multiplicities.

Reading

B. Hall, *Lie Groups, Lie Algebras, and Representations*, second edition (2015), Sections 10.7–10.8; Hall uses a convex-hull argument where we use dominance. J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Section 24 and its appendix. R. Carter, *Lie Algebras of Finite and Affine Type* (2005), Chapter 12. Exercise Sheet 16 repeats the proof in rank one.